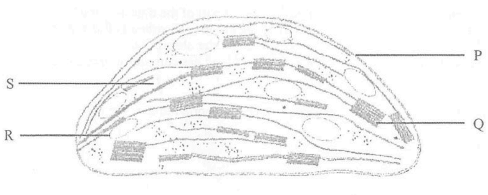
1. 2012Q2
Plants carry out photosynthesis to produce glucose which is required for the formation of
(1) cellulose.
(2) protein.
(3) starch.
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2. 2012Q23
Which of the following descriptions about the role of light in photosynthesis are correct?
(1) Activation of chlorophyll provides high energy electrons.
(2) Photolysis of water releases oxygen for use in carbon fixation.
(3) Photolysis of water releases hydrogen for the formation of NADPH.
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

3. 2013Q8
Which of the following processes in photosynthesis require energy input from ATP?
(1) fixation of carbon dioxide and the formation of 3-C compound
(2) reduction of 3-C compound leading to the formation of glucose
(3) regeneration of carbon dioxide acceptor
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

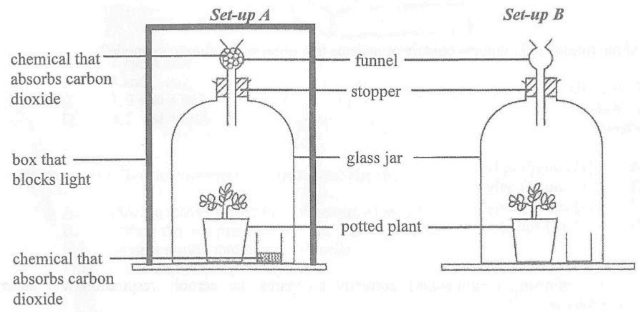
4. 2015Q4
Question 4 refers to the schematic diagram below, which shows the structures of a chloroplast:



Regeneration of the carbon dioxide acceptor takes place at
A. P.
B. Q.
C. R.
D. S.

Bio - By topic - Topic 18 Photosynthesis (2012-2024 version) by **DSERESO**

5. 2015Q11
Questions 11 and 12 refer to the following experiment. A student put two similar plants in darkness for 24 hours and then placed them in the following set-ups to conduct an investigation on photosynthesis:



At the end of the experiment, leaves were taken from the plants in set-ups A and B for the iodine test. Arrange the following steps in the correct order:
(1) Put the leaf in boiling water for 5 minutes.
(2) Add iodine solution to the leaf.
(3) Put the leaf in hot alcohol solution for 5 minutes.
(4) Put the leaf in water at room temperature for a few seconds.
A. (1), (2), (3), (4)
B. (1), (3), (4), (2)
C. (2), (3). (4), (1)
D. (4), (3), (2), (1)

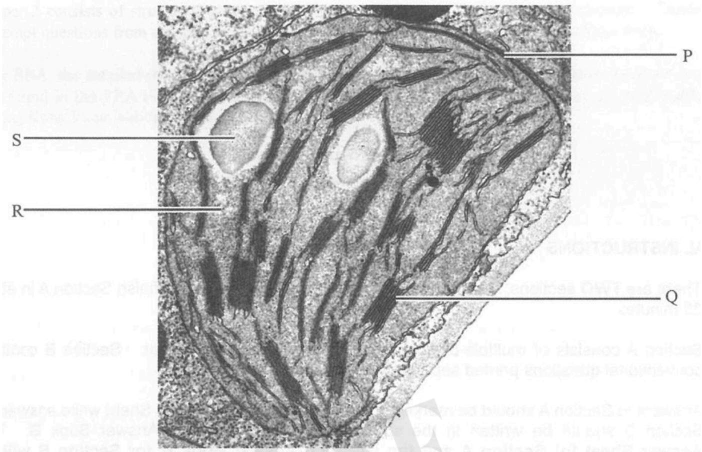
6. 2015Q12
After the iodine test, the leaf taken from set-up A was brown while the leaf taken from set-up B was blue-black. Which of the following conclusions can be drawn from the results?
A. Light is necessary for photosynthesis.
B. Carbon dioxide is necessary for photosynthesis.
C. Both light and carbon dioxide are necessary for photosynthesis.
D. Photosynthesis occurs in the plant in set-up B but not in set-up A.

7. 2017Q6
Which of the following reactions in photosynthesis takes place on the thylakoid membrane?
A. regeneration of carbon dioxide acceptor
B. reduction of 3-C compound
C. photolysis of water
D. carbon dioxide fixation

8. 2018Q1
Which of the following processes involves enzymes on cell membranes?
A. excretion of carbon dioxide by the lungs
B. transport of water along the xylem vessel
C. Calvin cycle in the chloroplasts of plant cells.
D. digestion of carbohydrates in the small intestine.

9. 2019Q1

Questions 1 and 2 refer to the electron micrograph below, which shows a chloroplast of a plant cell:



During photosynthesis, light is captured at

A. P.
B. Q.
C. R.
D. S.

10. 2019Q2

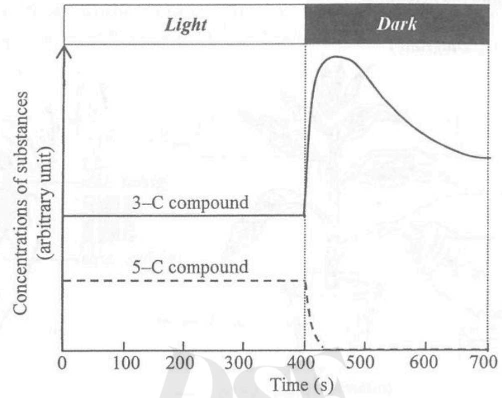
Carbon dioxide with radioactively labelled oxygen was provided to the plant cell for photosynthesis. Radioactivity can be detected in

A. oxygen produced at Q.
B. oxygen produced at R.
C. glucose produced at Q.
D. glucose produced at R.

Bio - By topic - Topic 18 Photosynthesis (2012-2024 version) by **DSERESO**

11. 2019Q4

The graph below shows the changes in the relative concentrations of a 3-C compound and a 5-C compound (carbon dioxide acceptor) in the Calvin cycle in green plant cells which have been kept in bright light and then in darkness

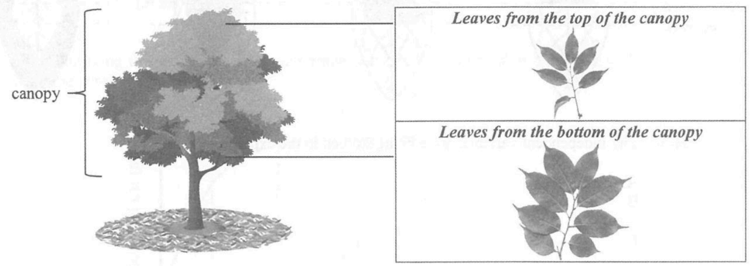


Which of the following is ***not*** a reason why the concentration of the 5-C compound decreased in the dark?

A. The 5-C compound was converted to the 3-C compound.
B. The 5-C compound combined with carbon dioxide to form glucose directly.
C. Regeneration of the 5-C compound stopped because there was no ATP from photochemical reactions.
D. Regeneration of the 5-C compound stopped because there was no NADPH from photochemical reactions.

12. 2019Q5

The photographs on the right below show leaves taken from different parts of the canopy of the same tree. (Note: The photographs are of the same magnification.)

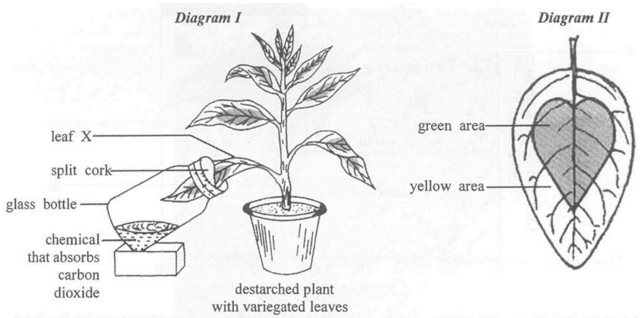


Which of the following is the most likely explanation for the differences between the leaves taken from the two parts of the canopy?

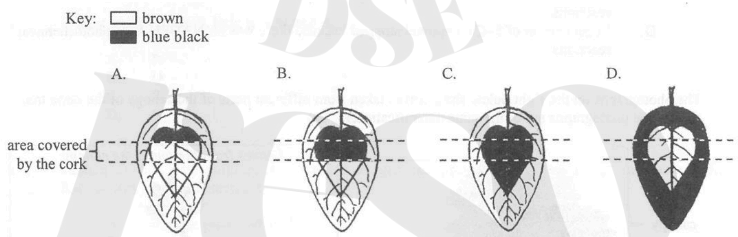
A. The leaves from the top of the canopy are smaller because they do not receive sufficient water for growth.
B. The leaves from the top of the canopy are smaller because they can reduce water loss due to transpiration.
C. The leaves from the bottom of the canopy are larger because they can store more food from photosynthesis.
D. The leaves from the bottom of the canopy are larger because they can collect light escaped through the top of the canopy.

13. 2019Q6

Questions 6 and 7 refer to Diagram I and Diagram II below. Diagram I shows a set-up prepared by a student to study the conditions for photosynthesis. Diagram II shows the leaf surface of a variegated leaf X before the experiment.



After leaving the set-up under sunlight for several hours, iodine test was carried out on leaf X. Which of the following diagrams correctly shows the results?



14. 2019Q7

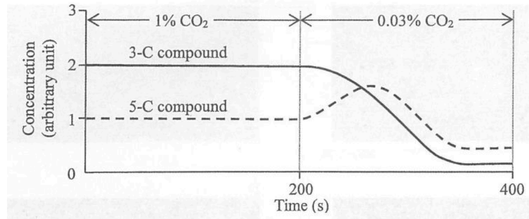
How many independent variables were being studied in the experiment

A. 1
B. 2
C. 3
D. 4

Bio - By topic - Topic 18 Photosynthesis (2012-2024 version) by **DSERESO**

15. 2022Q26

Questions 26 and 27 refer to an experiment in which a green algal culture was supplied with 1% carbon dioxide for 200 s and then followed by 0.03% carbon dioxide for another 200 s. The changes in the relative concentrations of a 3-C compound and a 5-C compound in the Calvin cycle of the green algae are shown below:



When the carbon dioxide concentration switched from 1% to 0.03%, which of the following combinations correctly shows the initial change of the concentration of the compound and its explanation?

	Initial change in concentration	***Explanation***
A.	3-C compound decreases	reduction of 3-C compound has increased
B.	3-C compound decreases	ATP from photochemical reactions has decreased
C.	5-C compound increases	carbon fixation has decreased
D.	5-C compound increases	regeneration of carbon dioxide acceptor has increased

16. 2022Q27

Which of the following factors should be kept constant throughout the experiment

(1) pH
(2) temperature
(3) light intensity

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

17. 2024Q7

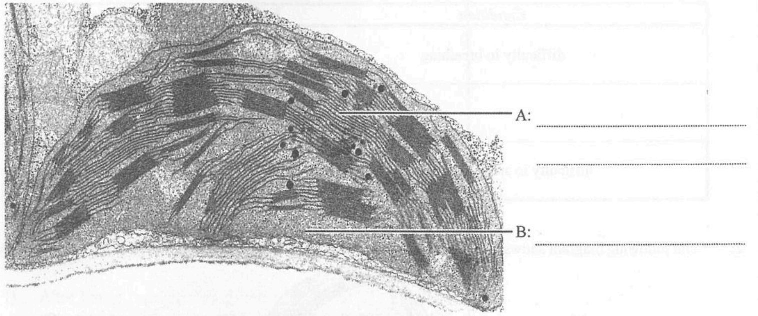
Which of the following combinations about the stages of photosynthesis is correct?

	Photochemical reactions	***Calvin cycle***
A.	produces oxygen	produces water
B.	requires carbon dioxide	requires NADPH
C.	occurs in stroma	occurs in grana
D.	light is the energy source	ATP is the energy source

LQ

1. 2016Q3

The diagram below shows the electron micrograph of an organelle:



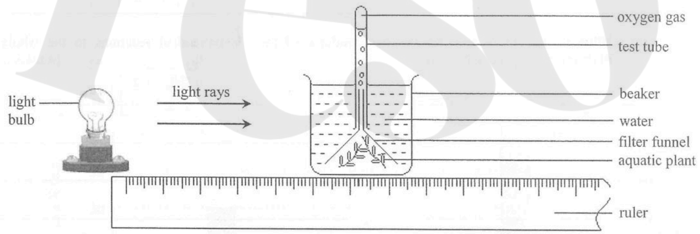
a) Label A and B. (2 marks)

b) State a type of plant cell that contains this organelle. (1 mark)

c) What is the functional relationship between A and B? (3 marks)

2. 2017Q7

The diagram below shows an experimental set-up for investigating the effect of light intensity on the rate of photosynthesis:



a) What is the assumption behind using the volume of oxygen released per unit time to indicate the photosynthetic rate? Explain your answer. (2 marks)

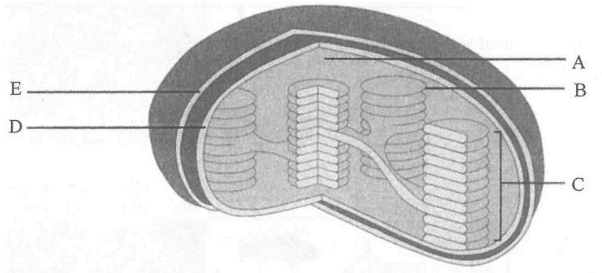
b) Suggest ***one*** modification to this experimental set-up to make sure that the result is due to the independent variable only. Explain your answer. (3 marks)

c) What is the significance of the ***two*** products of the photochemical reactions to the whole photosynthetic process? (4 marks)

Bio - By topic - Topic 18 Photosynthesis (2012-2024 version) by **DSERESO**

3. 2020Q1

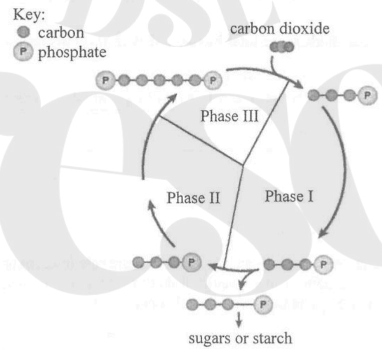
The diagram below shows the structures in a chloroplast:



a) Using the letters from the diagram, list ***all*** of the structures that contain photosynthetic pigments. (1 mark)

b) Structure C produces intermediates that are used in the Calvin Cycle. State the intermediates. (1 mark)

c) The diagram below shows a simplified Calvin Cycle:



Match the three phases with the following reactions: (2 marks)

Reactions	***Phase***
Regeneration of carbon dioxide acceptor	____________
Reduction of 3C compound	____________
Carbon dioxide fixation	____________

4. 2021Q8a, b

The table below shows the average blade area, blade thickness and thickness of the palisade mesophyll of leaves collected from the upper and lower regions of a tree species:

Location of leaves	Average blade area (cm^2)	Average blade thickness (μm)	Average thickness of palisade mesophyll (μm)
Upper region	62	177	45
Lower region	72	152	33

a) Compare the average blade area of leaves from the upper region and that from the lower region. With respect to the difference in surface area, suggest ***one*** adaptive advantage of the leaves from the lower region. (2 marks)

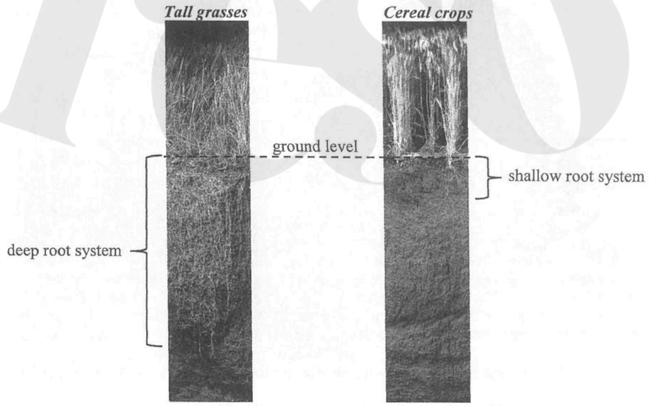
b) i) Compare the average thickness of the palisade mesophyll of leaves from the upper region and that from the lower region. (1 mark)

ii) Between the two types of leaves, suggest ***one*** possible structural difference which would lead to the difference stated in (bi). (1 mark)

iii) How would you confirm your answer in (bii)? (2 marks)

5. 2022Q8aii

Tall grasses and cereal crops belong to the same family and share a common ancestor. Cereal crops have been artificially selected for agriculture and their seeds harvested as food. The roots of the tall grasses in grasslands range in depth from 1.5 m to 4.5 m while those of cereal crops rarely exceed 1 m. The photographs below show the root depths of tall grasses and cereal crops under the same magnification:

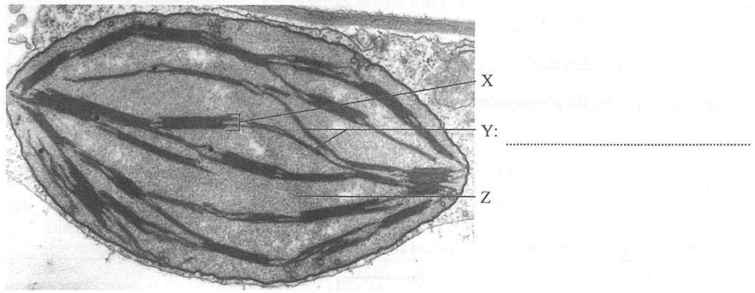


a) ii) In terms of energy usage of the plant, explain why having a shallow root system for cereal crops is considered an advantage to farmers. (2 marks)

Bio - By topic - Topic 18 Photosynthesis (2012-2024 version) by **DSERESO**

6. 2023Q5a, b, c

An electron micrograph of a chloroplast is shown below:



a) Label structure Y. (1 mark)

b) State the energy conversion which takes place at X and its importance in photosynthesis. (2 marks)

c) To which type of metabolism does the overall reaction at Z belong? Explain your answer. (2 marks)

Ans

2012Q2: D (33%)	2012Q23: B (68%)	2013Q8: C (42%)	2015Q4: D (57%)
2015Q11: B (96%)	2015Q12: D (60%)	2017Q6: C (73%)	2018Q1: D (71%)
2019Q1: B (83%)	2019Q2: D (44%)	2019Q4: B (49%)	2019Q5: D (45%)
2019Q6: A (77%)	2019Q7: C (31%)	2022Q26: C (44%)	2022Q27: D (62%)
2024Q7: D (78%)			

2016Q3

3 (a) thylakoid membrane (1)
stroma (1) (2)

(b) • mesophyll cell / palisade mesophyll cell / spongy mesophyll cell / guard cell (1) (1)

(c) • photochemical reactions take place at A (1)
• which supplies ATP and NADPH (1)
• for carbon fixation that takes place at B (1)
OR (3)
• carbon fixation takes place at B (1)
• which regenerate NADP (1)
• for photochemical reactions that take place at A (1)

6 marks

2017Q7

5 (a) • the rate of respiration remains constant throughout the experiment (1)
• so that any change in the net production of oxygen can be attributed to the change in the photosynthetic rate (1) (2)

(b) • place a thermometer in the beaker (1)
• to ensure the temperature is constant (1)
• as temperature is also a factor that may affect the photosynthetic rate (1)
OR (3)
• add a heat shield / water bath between the light bulb and the beaker / use a cold light source to conduct this experiment (1)
• to avoid the heating up of water in the beaker (1)
• as temperature is also a factor that may affect the photosynthetic rate (1)

(c) • photochemical reactions produce ATP (1)
• which provides energy to drive the light-independent reactions / for the regeneration of CO_2 acceptor (1)
• photochemical reactions also produce NADPH (1 (4)
• which provides reducing power for the reduction of 3-C compound to form glucose (1)

9 marks

2020Q1

1 (a) B and C (1) (1)

(b) ATP and NADPH / $NADPH^+$ / $NADPH_2$ / reduced NAD (1) (1)

(c) Phase II
Phase I
Phase III (2)
(Deduct 1 mark for each mistake)

4 marks

Bio - By topic - Topic 18 Photosynthesis (2012-2024 version) by **DSERESO**

2021Q8a, b

8 (a) • leaves taken from the lower region have a larger leaf area than those from the upper region (1) (2)
• a larger surface area increases the chance of capturing light (1)

(b) (i) • leaves taken from the upper region have a thicker palisade mesophyll than those taken from the lower region (1) (1)

(ii) • longer palisade cells / additional layers of palisade cells (1) (1)

(iii) • observe a cut section of the leaf under the microscope (1)
• measure the length or size of the palisade mesophyll cell / count the number of layers of palisade mesophyll cells (1) (2)

2022Q8aii

8 (a) (ii) • more energy can be allocated to the development of the shoot / less energy is needed for the development of the root (1)
• a better developed shoot will have a higher photosynthetic rate / support better development of seeds and fruits / a higher yield will be obtained (1) with the same amount of energy input (2)

2023Q5a, b, c

5 (a) • thylakoid / thylakoid membrane (1) (1)

(b) Concept for mark award:
• energy conversion at X (1)
• significance of the energy conversion in photosynthesis (1) (2)

e.g.
• light energy is captured in X and converted to chemical energy (1)
• to provide energy in the form of NADPH and ATP for Calvin cycle / dark reactions / carbon fixation (1)

(b) Concept for mark award:
• correct identification of the type of metabolism (1)
• example of the usage of energy to construct large molecules from small molecules (1) (2)

e.g.
• they are anabolic in nature / they are anabolic reactions / they belong to anabolism (1)
• using the energy captured to produce 3-C compounds / triose phosphate / glucose / starch / carbohydrates from carbon dioxide (1-C) (1)